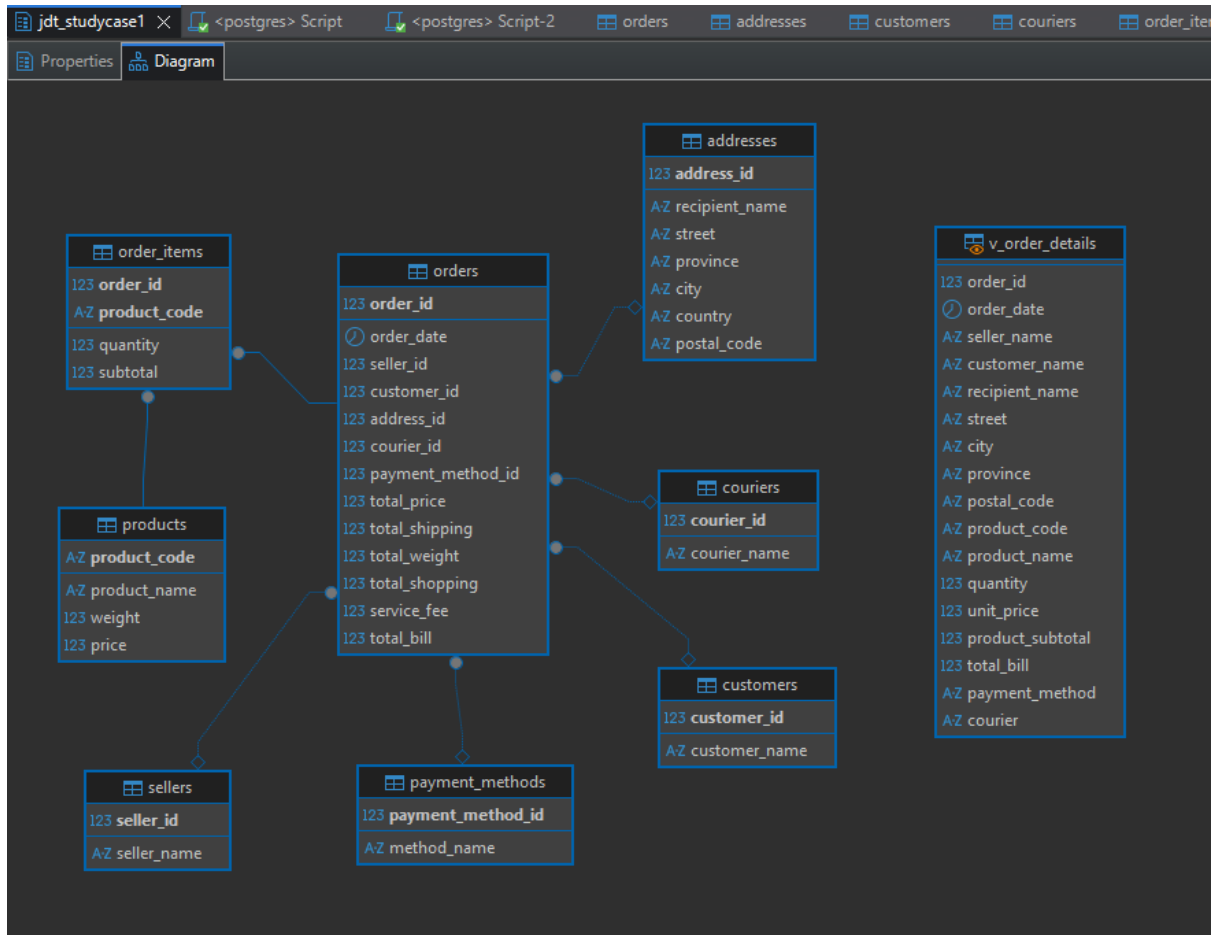


Nama: Julius Wahyu Wiccaksono

Tugas Normalisasi 3NF

ERD dan SQL

Gambar ERD dari dbeaver



Code Block SQL

Pembuatan schema dan table, dengan insert data.

```
CREATE SCHEMA IF NOT EXISTS JDT_studyCase1;
SET search_path TO JDT_studyCase1;

CREATE TABLE sellers (
    seller_id SERIAL PRIMARY KEY,
    seller_name VARCHAR(255) NOT NULL UNIQUE
);

CREATE TABLE customers (
    customer_id SERIAL PRIMARY KEY,
    customer_name VARCHAR(255) NOT NULL UNIQUE
);

CREATE TABLE couriers (
    courier_id SERIAL PRIMARY KEY,
    courier_name VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE payment_methods (
    payment_method_id SERIAL PRIMARY KEY,
    method_name VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE products (
    product_code VARCHAR(50) PRIMARY KEY,
    product_name VARCHAR(255) NOT NULL,
    weight NUMERIC,
    price NUMERIC
);

CREATE TABLE addresses (
    address_id SERIAL PRIMARY KEY,
    recipient_name VARCHAR(255),
    street TEXT,
    province VARCHAR(100),
    city VARCHAR(100),
    country VARCHAR(100),
    postal_code VARCHAR(20)
);

CREATE TABLE orders (
    order_id BIGINT PRIMARY KEY,
    order_date DATE,
    seller_id INT REFERENCES sellers(seller_id),
    customer_id INT REFERENCES customers(customer_id),
    address_id INT REFERENCES addresses(address_id),
    courier_id INT REFERENCES couriers(courier_id),
    payment_method_id INT REFERENCES payment_methods(payment_method_id),
    total_price NUMERIC,
    total_shipping NUMERIC,
    total_weight NUMERIC,
    total_shipping NUMERIC,
    service_fee NUMERIC,
    total_bill NUMERIC
);

CREATE TABLE order_items (
    order_id BIGINT REFERENCES orders(order_id),
    product_code VARCHAR(50) REFERENCES products(product_code),
    quantity NUMERIC,
    subtotal NUMERIC,
    PRIMARY KEY (order_id, product_code)
);
```

```

INSERT INTO sellers (seller_id, seller_name) VALUES (1, 'Galeri Olahraga');
INSERT INTO sellers (seller_id, seller_name) VALUES (2, 'Hello Store');

INSERT INTO customers (customer_id, customer_name) VALUES (1, 'Joko Morro');
INSERT INTO customers (customer_id, customer_name) VALUES (2, 'Eko Kurniawan');

INSERT INTO couriers (courier_id, courier_name) VALUES (1, 'Sicepat - Gokil');
INSERT INTO couriers (courier_id, courier_name) VALUES (2, 'JNE');

INSERT INTO payment_methods (payment_method_id, method_name) VALUES (1, 'Debit Online');
INSERT INTO payment_methods (payment_method_id, method_name) VALUES (2, 'Credit Card');

INSERT INTO addresses (address_id, recipient_name, street, province, city, country, postal_code)
VALUES (1, 'Budi Nugraha', 'Jalan raya xxx', 'DKI Jakarta', 'Jakarta', 'Indonesia', '433333');
INSERT INTO addresses (address_id, recipient_name, street, province, city, country, postal_code)
VALUES (2, 'Rully', 'Jalan raya baru', 'Jawa Barat', 'Bandung', 'Indonesia', '432434');

INSERT INTO products (product_code, product_name, weight, price) VALUES ('P0001', 'Bola Basket Size
7', 2300, 177900);
INSERT INTO products (product_code, product_name, weight, price) VALUES ('P0005', 'Apple iPhone Pro
Max', 1000, 25000000);
INSERT INTO products (product_code, product_name, weight, price) VALUES ('P0002', 'Bola Basket Size
5', 500.0, 98900.0);
INSERT INTO products (product_code, product_name, weight, price) VALUES ('P0006', 'Apple Watch 8',
2000.0, 8000000.0);
INSERT INTO products (product_code, product_name, weight, price) VALUES ('P0003', 'Pentil Pompa
Bola', 54.0, 9900.0);
INSERT INTO products (product_code, product_name, weight, price) VALUES ('P0004', 'Pompa Bola',
198.0, 43900.0);

INSERT INTO orders (order_id, order_date, seller_id, customer_id, address_id, courier_id,
payment_method_id, total_price, total_shipping, total_weight, total_shipping, service_fee,
total_bill) VALUES (12345, '2023-04-11', 1, 1, 1, 1, 1, 508500, 60000, 5360, 568500, 1000, 569500);
INSERT INTO orders (order_id, order_date, seller_id, customer_id, address_id, courier_id,
payment_method_id, total_price, total_shipping, total_weight, total_shipping, service_fee,
total_bill) VALUES (11111, '2023-04-12', 2, 2, 2, 2, 2, 33000000, 100000, 2000, 33100000, 1000,
33101000);
INSERT INTO orders (order_id, order_date, seller_id, customer_id, address_id, courier_id,
payment_method_id, total_price, total_shipping, total_weight, total_shipping, service_fee,
total_bill) VALUES (22222, '2023-04-13', 1, 1, 1, 1, 1, 355800, 60000, 2300, 415800, 1000, 416800);

INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (12345, 'P0001', 2,
355800);
INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (12345, 'P0002', 1.0,
98900.0);
INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (12345, 'P0003', 1.0,
9900.0);
INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (12345, 'P0004', 1.0,
43900.0);
INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (11111, 'P0005', 1,
25000000);
INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (11111, 'P0006', 1.0,
8000000.0);
INSERT INTO order_items (order_id, product_code, quantity, subtotal) VALUES (22222, 'P0001', 2,
355800);

```

Pembuatan view *v_order_details*.

```
SET search_path TO JDT_studycase1;

CREATE OR REPLACE VIEW v_order_details AS
SELECT
    o.order_id,
    o.order_date,
    s.seller_name,
    c.customer_name,
    a.recipient_name,
    a.street,
    a.city,
    a.province,
    a.postal_code,
    p.product_code,
    p.product_name,
    oi.quantity,
    p.price AS unit_price,
    oi.subtotal AS product_subtotal,
    o.total_bill,
    pm.method_name AS payment_method,
    cr.courier_name AS courier
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_code = p.product_code
JOIN sellers s ON o.seller_id = s.seller_id
JOIN customers c ON o.customer_id = c.customer_id
JOIN addresses a ON o.address_id = a.address_id
JOIN payment_methods pm ON o.payment_method_id = pm.payment_method_id
JOIN couriers cr ON o.courier_id = cr.courier_id;
```